START FROM SCRATCH / HYPER-PACED LEARNING

Week 5: Explore multi-agent systems

Give the SRE Agent specialist partners only when the work benefits from delegation, parallel investigation, or independent review.

WHAT YOU ALREADY HAVE

One bounded SRE Agent

The SRE Assistant can use runbook retrieval and safe tools in a visible, bounded loop. Week 4 established the single-agent baseline to compare against.

THIS WEEK'S PLAN

Coordinate specialists with intent

Identify a task that can genuinely be split. Define a coordinator, specialist responsibilities, handoffs, shared evidence, and how a final answer is assembled.

RED FLAGS / TESTING

Prove the extra coordination pays off

Check delegated-task quality, duplicate or conflicting work, failed handoffs, cost, latency, and stop conditions. Compare against the single-agent baseline; do not add agents just to make the diagram look advanced.

EXPECTED OUTCOME

A justified multi-agent design

Show one coordinated SRE investigation and explain what improved, what became harder, and whether the team should keep the multi-agent design or return to one agent.

More agents do not automatically mean a better system. Make coordination earn its complexity.

Team rhythm: prepare independently, align with your team and lead, build together, then share. Each team has 15 minutes: 5 minutes to demo and 10 minutes for discussion and feedback.

PREPARE BEFORE YOU BUILD

Understand delegation before adding agents

Focus on the problem shape, task boundaries, and coordination costs. You do not need to adopt a specific framework to understand the patterns.

01 **Building Effective AI Agents**
Anthropic Engineering
Read the sections on parallelization and orchestrator-workers. Note when tasks are independent and when a central agent needs to decide what work to delegate.
Read the patterns

02 **How we built our multi-agent research system**
Anthropic Engineering
Study the coordinator and parallel specialist pattern, then note the operational lessons: agents need bounded tasks, coordination, evaluation, and clear stopping rules.
Read the case study

Bring these questions to your team

- Which SRE investigation can be split into independent tasks without losing shared context?
- What evidence must each specialist return so the coordinator can compare and synthesize it?
- What measurable result would justify the extra latency, cost, and coordination complexity?

The Week 5 goal is not to build a large agent swarm. Start with one coordinator and a small number of clear specialist roles, or conclude that one agent is sufficient.

DESIGN THE TEAM, NOT A SWARM

A simple delegation pattern

A multi-agent design is a collaboration pattern around a task. Make the interfaces explicit so the team can see what is delegated, what comes back, and who owns the final answer.

1 / Coordinator	2 / Specialists	3 / Synthesis
Clarifies the incident goal, chooses which specialists to call, and decides whether evidence is sufficient.	Investigate bounded questions, such as service health, relevant runbook guidance, or recent change context.	Checks returned evidence for gaps or conflicts, then produces one concise recommendation or escalation.

Define each handoff

Handoff	What must be clear	Failure to test
Coordinator -> specialist	Task, scope, available tools, evidence requested, time or call limit	Vague task or unnecessary delegation
Specialist -> coordinator	Finding, source, confidence or uncertainty, errors, next question	Unsupported claim or empty result
Coordinator -> user	What was checked, what remains uncertain, recommendation, escalation	Contradictory results hidden in a polished summary

Keep the Week 4 baseline

Run the same incident examples through the original single SRE Agent and the multi-agent version. Compare answer usefulness, evidence coverage, completion time, tool calls, and operational complexity. A multi-agent system is a win only when the evidence supports it.

COME READY TO DESIGN

Your Multi-Agent Design Canvas

Use this before implementation. Keep the roles small, the handoffs visible, and the single-agent comparison available.

1. Task What SRE outcome are we trying to achieve?	**2. Split** Which independent subtasks justify delegation?
3. Roles What does each specialist own, and what can it not do?	**4. Handoff** What structured evidence does each specialist return?
5. Coordination How are conflicts, missing results, or timeouts handled?	**6. Stop rule** When does the coordinator stop calling specialists?
7. Baseline How will this compare with one agent?	**8. Demo** What visible evidence proves the team improved the result?

Show the learning

- Show the same incident handled by the single-agent baseline and by the coordinated version.
- Show a failed or conflicting specialist result and how the coordinator handles it.
- Explain the trade-off: quality or coverage gained versus added cost, latency, and complexity.

Keep the problem statement short. Show the delegation and evidence. Present, discuss, and learn from every team.